MODULE: INFORMATION TRANSFER

DNA as the genetic material:

The unequivocal proof that DNA is the genetic material came from the experiments of Alfred Hershey and Martha Chase (1952). They worked with viruses that infect bacteria called bacteriophages.

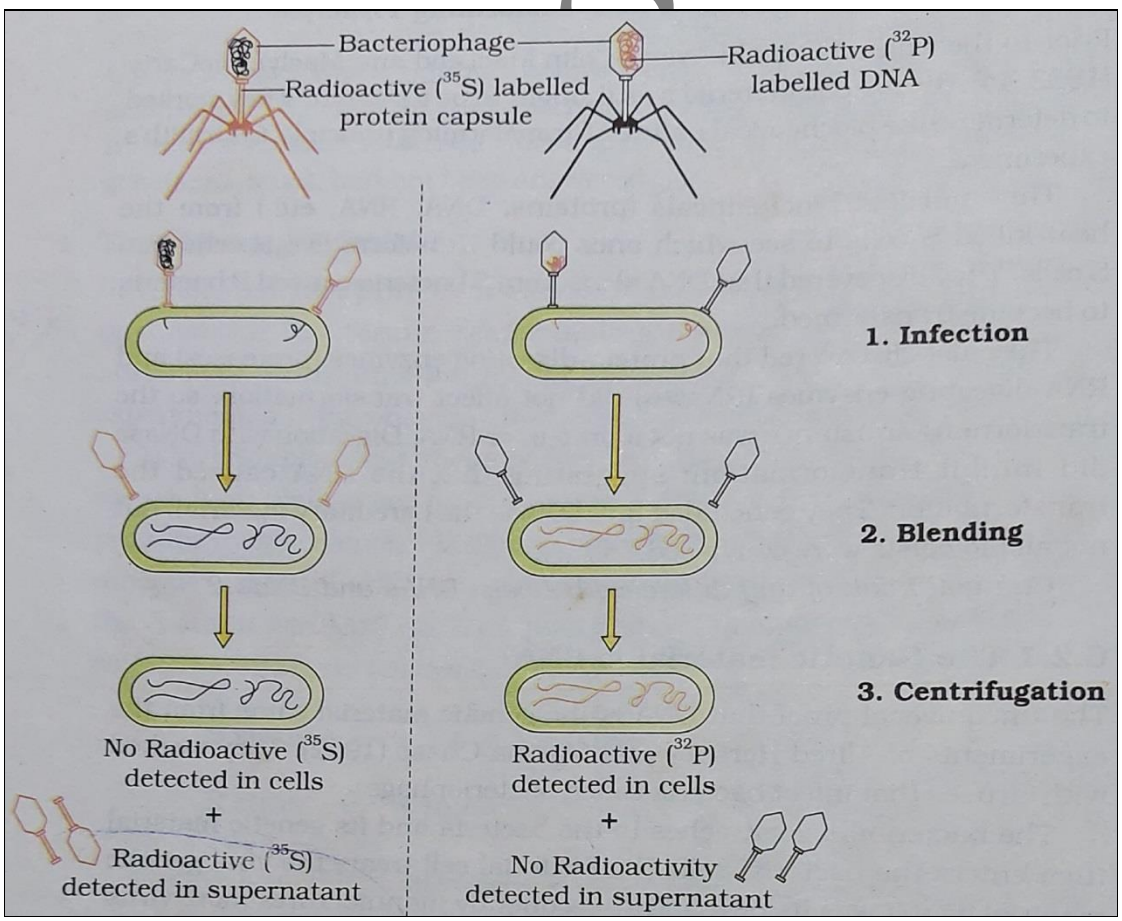
The bacteriophage attaches to the bacteria and its genetic material then enters the bacterial cell. The bacterial cell treats the viral genetic material as if it was its own and subsequently manufactures more virus particles. Hershey and Chase worked to discover whether it was protein or DNA from the viruses that entered the bacteria.

They grew some viruses on a medium that contained radioactive phosphorus and some others on medium that contained radioactive sulphur. Viruses grown in the presence of radioactive phosphorus contained radioactive DNA but not radioactive protein because DNA contains phosphorus but protein does not. Similarly, viruses grown on radioactive sulphur contained radioactive protein but not radioactive DNA because DNA does not contain sulphur.

Radioactive phages were allowed to attach to *Escherichia coli* bacteria. Then, as the infection proceeded, the viral coats were removed from the bacteria by agitating them in a blender. The virus particles were separated from the bacteria by spinning them in a centrifuge.

Bacteria which were infected with viruses that had radioactive DNA were radioactive, indicating that DNA was the material that passed from the virus to the bacteria. Bacteria that were infected with viruses that had radioactive proteins were not radioactive. This indicates that proteins did not enter the bacteria from the viruses. DNA is therefore the genetic material that is passed from virus to bacteria.

Singh



Hierarchy of DNA structure – from single stranded to double helix to nucleosome:

DNA is a nucleic acid that contains the genetic instructions used in the development and functioning of all known living organisms. The main role of DNA molecule is the long-term storage of information and is often compared to a set of blueprints, since it contains the instructions needed to construct other components of cells, such as RNA molecules and proteins. DNA is made up of four types of nucleotides, containing different nitrogenous bases, the pyrimidines (cytosine and thymine) and the purines (adenine and guanine). The nucleotides are attached to each other in a chain by bonds between their sugar and phosphate groups, forming a sugar-phosphate backbone. Two of these chains are held together by hydrogen bonding between complementary bases (adenine pairing with thymine via two hydrogen bonds and guanine pairing with cytosine via three hydrogen bonds). The chains coil around each other forming the DNA double helix.

In 1953, James Watson and Francis Crick based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin proposed a very simple but famous double helix model for the structure of DNA. The proposition of the model was also based on the observation of Erwin Chargaff that for a double stranded DNA; the ratios between purine and pyrimidine are constant and equal one. Salient features of the Double-helix structure of DNA are as follows:

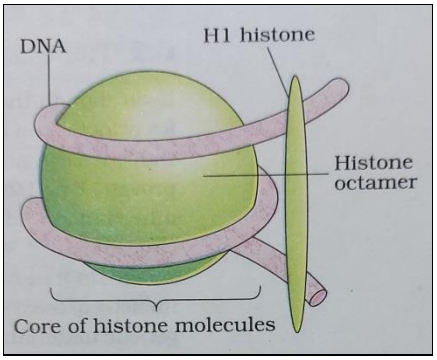
- It is made up of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases projects inside.
- The two chains have anti-parallel polarity. It means, if one chain has the polarity 5' → 3', the other has 3' → 5'.
- The bases in two strands are paired through hydrogen bond forming base pairs (bp).
- The DNA chain is approximately 20 Å wide.
- The two chains are coiled in a right-handed fashion.
- The pitch (on complete turn of the DNA) of the helix is 3.4 nm.
- There are roughly 10 bp in each turn.
- The distance between a base pair in a helix is approximately equal to 0.34 nm.

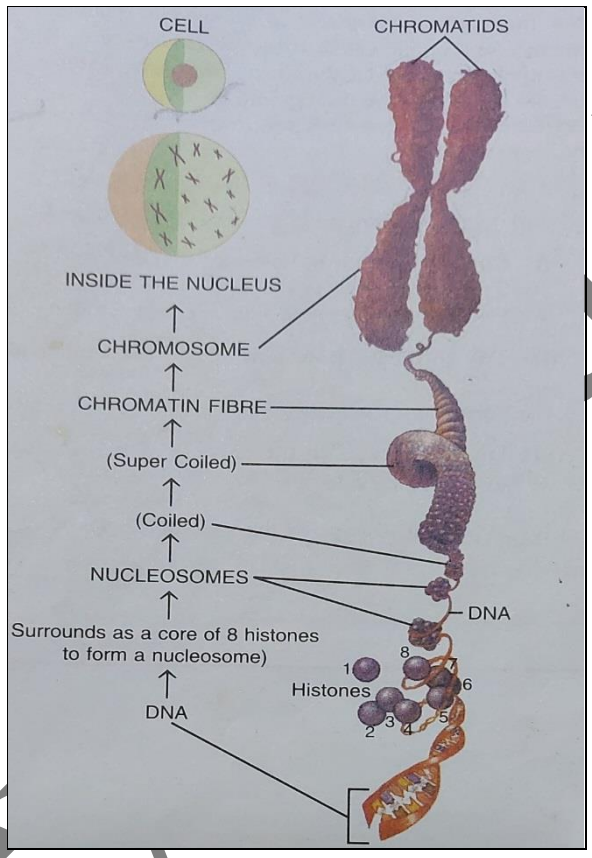
Taken the distance between two consecutive base pairs as 0.34 nm ($0.34X10^{-9}$m), if the length of DNA double helix ($6.6X10^{9}$bp) in a typical mammalian cell is calculated, it comes out to be ($6.6X10^{9}X0.34X10^{-9}$) approximately 2.2 metres. A length that is far greater than the dimension of a typical nucleus (approximately 10^{-6}m). Then the question that arises is – how is such a long polymer packaged in a cell?

In eukaryotes, this packaging or organisation is much complex. There is a set of positively charged, basic proteins called Histones. A protein acquires charge depending upon the abundance of amino acid residues with charged side chains. Histones are rich in the basic amino acids like lysine and arginine. Both the amino acids carry positive charges in their side chains. Histones are organised to form a unit of eight molecules called as histone octamer. The negatively charged DBNA is wrapped around the positively charged histone octamer to form a structure called Nucleosome. A typical nucleosome contains 200 bp of DNA helix. Nucleosomes constitute the repeating unit of a structure in nucleus called Chromatin, thread-like stained bodies seen in nucleus. The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope.

The beads-on-string structure in chromatin is packaged to form chromatin fibres that are further coiled and condensed at metaphase stage of cell division to form chromosomes. The packaging of chromatin at higher level requires additional set of proteins that collectively are referred to as Non-histone Chromosomal (NHC) proteins.

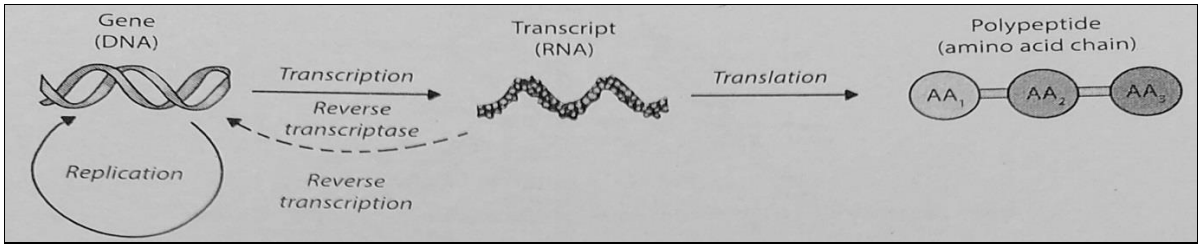
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The molecular basis of coding and decoding genetic information is universal:

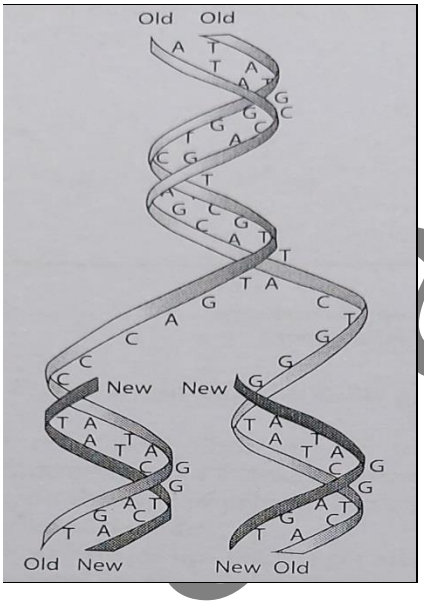
The Central Dogma in molecular biology proposed by Francis Crick stated that the flow of genetic information is universal and is from DNA → RNA → Protein.
In some viruses (retroviruses) the flow of information is in reverse direction. Herein RNA is used as a template to synthesize DNA. This process is called reverse transcription and the process is catalysed by the enzyme reverse transcriptase.



Molecular basis of information transfer:

A. Replication of DNA:

Watson and Crick suggested that replication would occur by gradual separation of the strands of the double helix. Since the two strands are complementary to each other, each strand carries the information required for the synthesis of the other strands. Hence, if the strands are separated, each can act as a template to direct the synthesis of the complementary strand. Since after replication, each molecule of DNA will have one old strand and one new strand, this type of replication is known as semi-conservative replication. DNA synthesis takes place only in 5' → 3' direction.

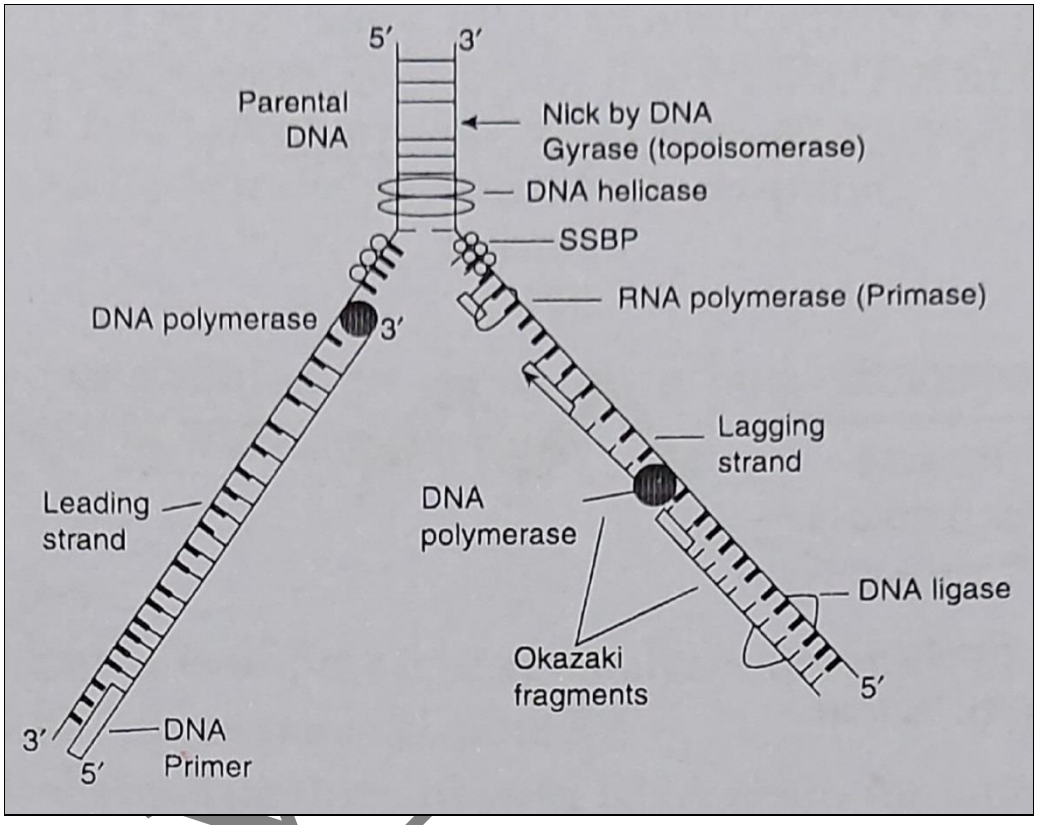


The entire process of DNA replication involves the following common steps:

1. Recognition of the initiation point:
   DNA helix unwinds by the enzyme helicase which use the energy of ATP and the replication of DNA begin at a specific point called point of origin ('ori' site / ARS).
2. Unwinding of DNA:
   The unwinding proteins bind to the nicked strand of the duplex and separate the two strands at DNA duplex. Topoisomerase helps in unwinding of DNA by relieving the topological stress created due to the helicase activity.
3. Single strand binding protein (SSBP):
   These proteins help in keeping DNA in single stranded position and hence also called helix destabilising protein.
4. RNA priming:
   The DNA directed RNA polymerase now synthesizes the primer strands of RNA (RNA primer), because the main polymerisation enzyme is incapable of initiating new strand synthesis. The RNA primers are generally 10 to 20 nucleotides long.
5. Primer extension / Polymerisation:
   The new strands of DNA are formed in the 5' → 3' direction from the 3' → 5' template DNA by the addition of deoxyribonucleotides to the 3' end of primer RNA. Addition of nucleotide is done by DNA polymerase III. The leading strand of DNA is synthesized continuously in 5' → 3' direction as one piece. The

lagging strand of DNA is synthesized discontinuously in its opposite direction in short segments. These segments are called Okazaki fragments.

6. Excision of RNA primers:
   Once Okazaki fragments have been formed, the RNA primers are removed from the 5' end by the action of 5' → 3' exonuclease activity of DNA polymerase I.
7. Joining of Okazaki fragments:
   The gaps left between Okazaki fragments are filled with complimentary deoxyribonucleotides residues by DNA polymerase I. Finally, the adjacent 5' and 3' ends are joined by DNA ligase.

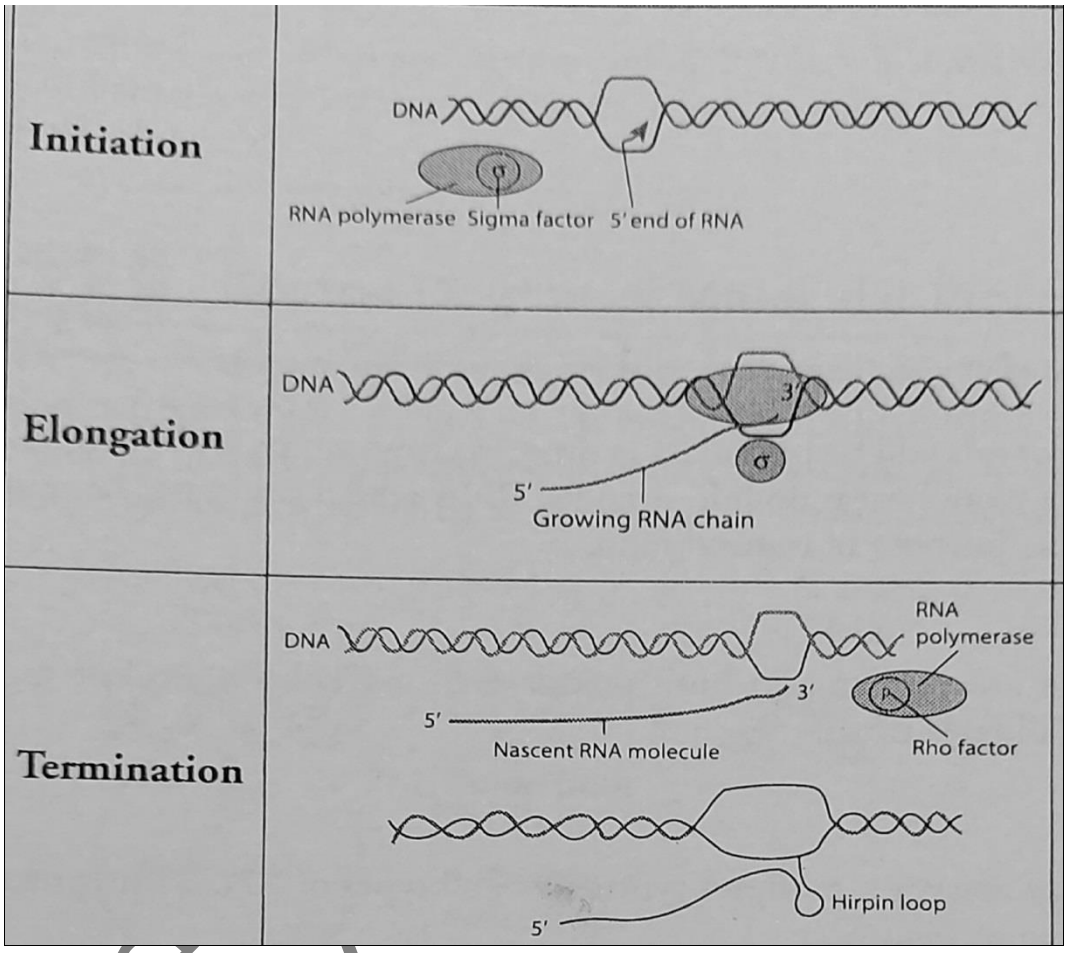


B. Transcription:

The process by which RNA is synthesized from its DNA template is known as transcription. It is generally considered as the first step in gene expression. Here also the principle of complementarity governs the process of transcription, except that the adenosine now forms base pair with uracil instead of thymine. However, unlike in the process of replication, which once set in, the total DNA of an organism gets duplicated, in transcription only a segment of DNA and only one of the strands is copied into RNA.

Why both the strands are not copied during transcription has the simple answer. First, if both strands act as template, they would code for RNA molecule with different sequences, and in turn, if they code for proteins, the sequence of amino acids in the proteins would be different. Hence, one segment of the DNA would be coding for two different proteins, and this would complicate the genetic information transfer machinery. Second, the two RNA molecules if produced simultaneously would be complementary to each other, hence would form a double stranded RNA. This would prevent RNA from being translated into protein and the exercise of transcription would become a futile one.

The mechanism of transcription is as follows: the DNA dependent RNA polymerase binds to promoter and initiates transcription (Initiation). It uses nucleoside triphosphates as substrate and polymerise in a template dependent fashion following the rule of complementarity. It somehow also facilitates opening of the helix and continues elongation. Only a short stretch of RNA remains bound to the enzyme. Once the polymerase reaches the terminator region, the nascent RNA falls off, so also the RNA polymerase. This results in the termination of transcription.



An intriguing question is that how is the RNA polymerase able to catalyse all the three steps, which are initiation, elongation and termination. The RNA polymerase is only capable of catalysing the progress of elongation. It associates transiently with initiation factor (σ factor) and termination factor (ρ factor) to initiate and terminate the transcription, respectively. Association with these factors alter the specificity of the RNA polymerase to either initiate or terminate.

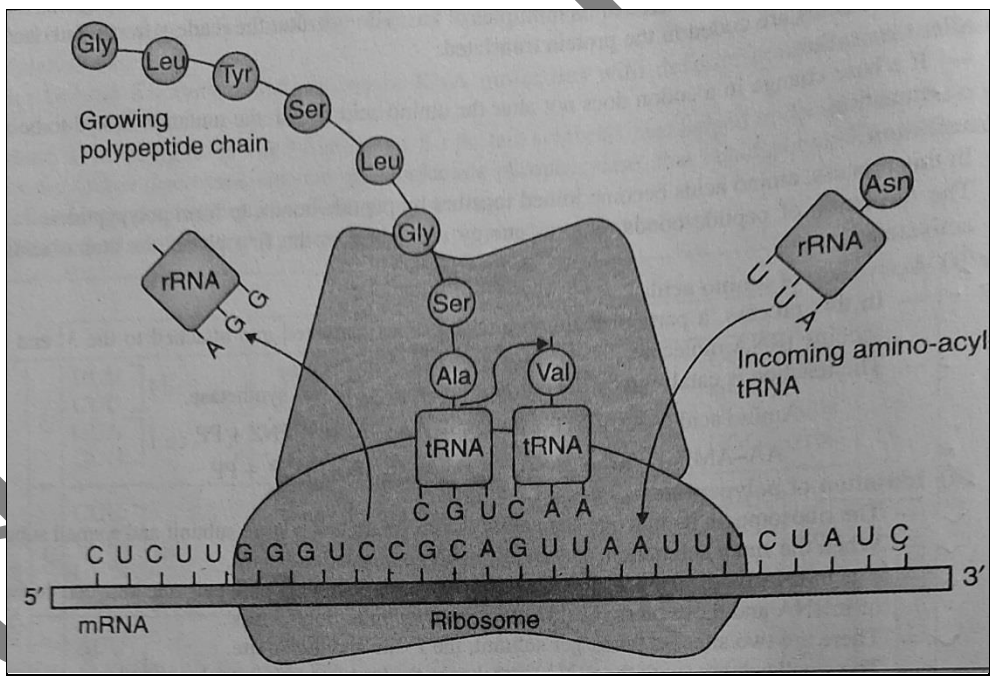
C. Translation:

Translation refers to the process of polymerisation of amino acids to form a polypeptide. The order and sequence of amino acids are defined by the sequence of bases in the mRNA. The amino acids are joined by a bond which is known as a peptide bond. Formation of a peptide bond requires energy. Therefore in the first phase itself amino acids are activated in the presence of ATP and linked to their cognate tRNA – a process commonly called as 'charging of tRNA' or 'aminoacylation of tRNA' to be more specific.

The cellular factory responsible for synthesizing proteins is the ribosome. The ribosome consists of structural RNAs (rRNA) and about 80 different proteins. In its inactive state, it exists as two subunits; a large subunit and a small subunit. When the small subunit encounters an mRNA, the process of translation of the mRNA to protein begins. There are two sites in the large subunit, for subsequent amino acids to bind to and thus, be close enough to each other for the formation of a peptide bond. The ribosome also acts as a catalyst for the formation of peptide bond.

A translational unit in mRNA is the sequence of RNA that is flanked by the start codon (AUG) and the stop codon (UAA/UGA/UAG) and codes for a polypeptide. An mRNA also has some additional sequences that are not translated and are referred as untranslated regions (UTR). The UTRs are present at both 5' end (before start codon) and at 3' end (after stop codon). They are required for efficient translation process.

For initiation, the ribosome binds to the mRNA at the start codon (AUG) that is recognised only by the initiator tRNA. The ribosome proceeds to the elongation phase of protein synthesis. During this stage, complexes composed of an amino acid linked to tRNA, subsequently bind to the appropriate codon in mRNA by forming complementary base pairs with the RNA codon. The ribosome moves from codon to codon along the mRNA. Amino acids are added one by one, translated into polypeptide sequences dictated by DNA and represented by mRNA. At the end, a release factor binds to the stop codon, terminating translation and releasing the complete polypeptide from the ribosome.

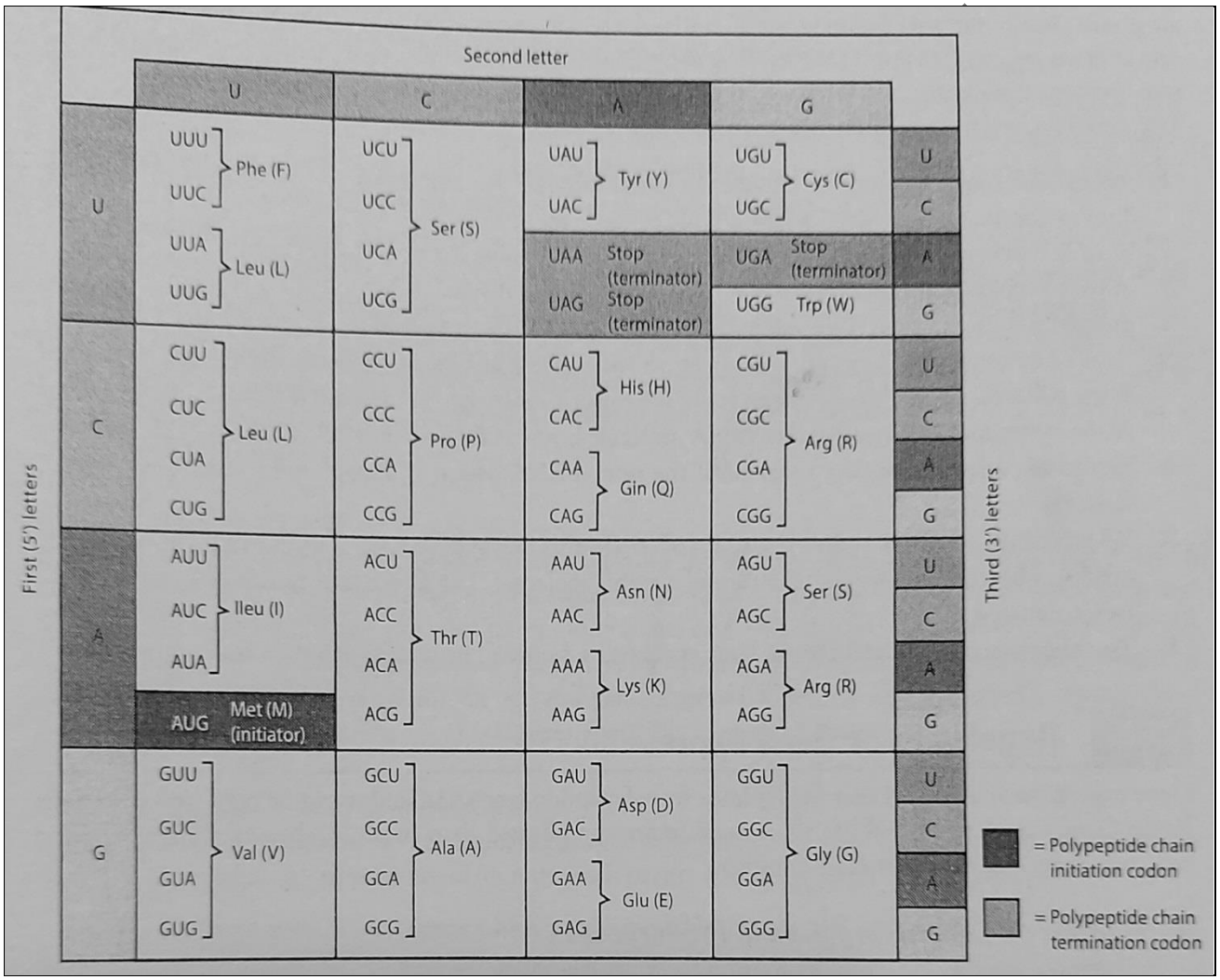


Concept of Genetic code. Universality and degeneracy of genetic code:

Genetic code is the genetic information present in the nucleotide sequences of DNA with three consecutive nucleotides functioning as a codon in a non-overlapping fashion. The genetic vocabulary consists of 64 codons. Each codon has a message for construction of polypeptide chain. Genetic code is, therefore, also defined as a correspondence between the sequence of amino acids in polypeptides and the nucleotide sequence in DNA.

Salient features of the genetic code are:

1. **Triplet:** A single amino acid is specified by a sequence of three nucleotides in mRNA that is called codon. Due to triplet nature, it consists of 64 codons.
2. **Universal:** A codon specifies the same amino acid in all organisms from viruses to human beings.
3. **Commaless:** There is no pause, so it reads continuously.
4. **Non-overlapping:** No overlapping between adjacent nucleotide.
5. Initiation codon: The synthesis of polypeptide chain initiated by initiation codon that is AUG that code for methionine amino acid.
6. **Termination codon:** Termination is done by terminator codons UAA, UGA or UAG which do not code for any amino acid. These are also called non-sense codon.
7. **Degenerate:** A single amino acid may be specified by many codons.



Define gene in terms of complementation and recombination:

A. Complementation:

In Genetics, complementation occurs when two strains of an organism with different homozygous recessive mutations that produce the same mutant phenotype have offspring that express the wild-type phenotype when mated or crossed.

Complementation will ordinarily occur if the mutations are in different genes (intergenic complementation). Complementation may also occur if the two mutations are at different sites within the same gene (intragenic complementation).

B. Recombination:

Recombination is a process by which pieces of DNA are broken and recombined to produce new combinations of alleles. This recombination process creates genetic diversity at the level of genes that reflects differences in the DNA sequences of different organisms.

In eukaryotic cells, which are cells with a nucleus and organelles, recombination typically occurs during meiosis. Meiosis is a form of cell division that produces gametes (egg or sperm cells). During the first phase of meiosis, the homologous pairs of maternal and paternal chromosomes align. During the alignment, the arms of the chromosome can overlap and temporarily fuse, causing a crossover. Crossovers result in recombination and the exchange of genetic material between the maternal and paternal chromosomes. As a result, offspring can have different combination of genes than their parents. Genes that are located farther apart on the same chromosome have a greater likelihood of undergoing recombination, which means they have a greater recombination frequency.

